

CARRIER-TO-BOARD INTERCONNECT DOSSIER

UNIVERSAL INTERFACE SPECIFICATION FOR HYBRID METALLO-VITREOUS MATRICES

STATUS: MAXIMAL-RESULT-PROTOCOL (HARDENED)

CLASSIFICATION: UNIVERSAL CARRIER TOPOLOGY & I/O STANDARDS

REVISION: 1.0 (UNABRIDGED INTERCONNECT SPECIFICATION)

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This document specifies the universal contact and transmission interfaces of the V5.0 Metallo-Vitreous Composite (MVC) architecture. The carrier acts as an agnostic, ultra-high-performance interposer substrate. It does not dictate silicon logic nor mainboard form factors (sockets). It exclusively standardizes the physical, electrical, and fluidic contact points required to bridge third-party active silicon with any industrial printed circuit board (PCB).

1. Strategic Abstraction: The Universal Carrier Concept

The MVC architecture isolates the thermal and electrodynamic complexity of modern high-end computing from the active silicon die. By functioning purely as a carrier, semiconductor manufacturers can utilize the matrix for any processing unit (CPU, GPU, NPU, or future material-based cognition units).

Design Paradigm:

- Top-Level (Die-to-Carrier):** Agnostic micro-bump arrays with thermal direct-contact zones.
- Bottom-Level (Carrier-to-Board):** Agnostic Under-Bump-Metallization (UBM) for custom BGA (Ball Grid Array) or LGA (Land Grid Array) implementations, alongside universal fluidic inlet/outlet ports.

2. Top-Level Interface: Die-to-Carrier Connection

The interface between the third-party silicon die and the MVC glass matrix utilizes a highly optimized Anisotropic Conductive Film (ACF) protocol, securing both electrical Z-axis conductivity and mechanical shear stress absorption.

2.1. Contact Resistance & Signal Integrity

The electrical contact points are structured via localized Aerosol Jet Printing (AJP) silver nanopads. The contact resistance (R_c) of a single Z-axis ACF particle connection is mathematically defined by the constriction resistance and film tunneling resistance:

$$R_c = \frac{\rho_{particle}}{2 \cdot a_{contact}} + \frac{\rho_{tunnel}}{A_{effective}}$$

Where $a_{contact}$ represents the radius of the deformed conductive particle under thermo-compression, and $A_{effective}$ is the active tunneling area. By scaling the ACF pad sizes to manufacturer specifications, R_c is reliably pushed below 10 mΩ per contact, guaranteeing loss-free signal input into the carrier.

2.2. Thermal Coupling Zone

The carrier omits standard Integrated Heat Spreaders (IHS). Instead, the silicon die sits directly above the internal 6-Bar HFE micro-fluidic channels. The interface requires only a highly conductive, non-curing liquid metal thermal interface material (TIM) or direct dielectric fluid contact at the defined thermal exclusion zones (TEZ).

3. Bottom-Level Interface: Carrier-to-Board I/O

The carrier connects to the mainboard without dictating a proprietary socket geometry. The manufacturer defines the footprint, while the carrier provides the standardized metallurgical and fluidic hardpoints.

3.1. Universal Under-Bump Metallization (UBM)

To ensure compatibility with all standard soldering processes (Lead-free SAC305, Bismuth alloys, or LGA gold-pads), the bottom TGVs (Through-Glass-Vias) of the carrier are terminated with a universal UBM stack: $Ti/Cu/Ni/Au$.

- **Titanium (Ti) Adhesion Layer:** 50 nm for covalent bonding to the glass matrix.
- **Copper (Cu) Conduction Layer:** 500 nm for bulk current transfer.
- **Nickel (Ni) Barrier Layer:** 3 μm to prevent copper diffusion into the solder joint.
- **Gold (Au) Oxidation Shield:** 50 nm flash layer to guarantee perfect wetting during manufacturer-side BGA ball drop or LGA socketing.

3.2. Universal Micro-Fluidic Manifold (UFM)

Instead of proprietary cooling blocks, the carrier features standardized high-pressure fluidic I/O ports. The flow resistance (ΔP) across the carrier-to-board transition must be minimized to maintain the 6-Bar system pressure. The pressure drop across the standardized circular inlet transition is:

$$\Delta P_{transition} = \zeta \cdot \frac{\rho \cdot v^2}{2}$$

Where $\zeta \approx 0.5$ (for an abrupt contraction/expansion at the O-ring interface), ρ is the density of the HFE fluid, and v is the flow velocity. The ports are sealed using standard aerospace-grade Viton micro-O-rings, allowing any hardware vendor to attach their own external pumping mechanisms.

4. High-Frequency & Inductive Contact Electrodynamics

4.1. Impedance Matching for Sub-THz I/O

For extreme data rates ($> 1,000$ Gbps/Lane), the transition from the carrier UBM to the PCB must maintain a strict characteristic impedance (Z_0). The design guidelines stipulate that the manufacturer must tune their PCB pad capacitance (C_{pad}) to match the carrier's internal transmission line inductance (L_{via}):

$$Z_0 = \sqrt{\frac{L_{via}}{C_{pad}}} \approx 50 \Omega$$

By providing exact parasitic values of the MVC carrier ($L_{via} \approx 0.1$ nH), third-party engineers can perfectly match their board layouts, eliminating signal reflections (Return Loss $S_{11} < -20$ dB).

4.2. Contactless Co2Z Control Bus

For control signals avoiding mechanical wear, the carrier exposes the planar Co2Z Hexaferrite micro-inserts at the bottom layer. The mainboard requires only a matching planar coil. The mutual inductance (M) across the physical air gap (g) is standardized:

$$M = k \cdot \sqrt{L_{carrier} \cdot L_{board}}$$

With the high-permeability inserts, the coupling factor stabilizes at $k > 0.90$ across a physical air gap of up to 2 mm, allowing completely socket-free, wear-resistant data transmission for system management tasks.

5. Form-Factor Agnosticism (Socket Independence)

By restricting the MVC architecture entirely to the role of a passive, ultra-high-performance transport and cooling medium, we eliminate vendor lock-in.

- The carrier can be laser-cut to 50×50 mm for server GPUs or 20×20 mm for mobile AI units.
- The contact arrays can be mapped dynamically during the IR-laser DLMS process.

- The responsibility for mechanical retention (clamping force) and the physical socket design is entirely outsourced to the ecosystem partners.

SECURITY AND EXPORT DECLARATION:

This specification is subject to the **Manual-Only Export principle** within the free ecosystem. It establishes the foundational integration rules for third-party hardware without relinquishing the sovereignty over the MVC matrix technology itself.